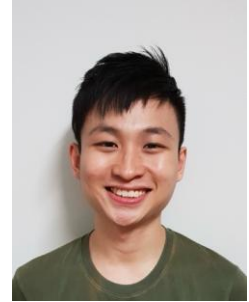


NICHOLAS **HANG** BAO HAN



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A clinician with a passion for public health, dedicated to addressing the intersection of clinical care, environmental and occupational health, and health systems. Committed to advancing evidence-based interventions and policies that promote health equity, safety, and sustainability for communities and workplaces.

EDUCATION

Doctor of Medicine (MD)

The University of Queensland (UQ) – Australia, Brisbane

Feb 2018 - Dec 2021

- Graduated with additional Certificate of Research Achievement.

Bachelor of Science (BSc) (Hons) in Biological Sciences

Aug 2012 - Aug 2016

Nanyang Technological University (NTU) – Singapore

- Graduated with 2nd Class Honours (Upper Division, Distinction)
- Honours thesis : Short environmental enrichment rescues effects of chronic stress in infralimbic neurons of medial prefrontal cortex (mPFC) in adult male rats.

Diploma in Environmental Water Technology

Apr 2007 - Oct 2010

Ngee Ann Polytechnic (NP) - Singapore

- Graduated with Certificate plus in Japanese Language.
- Final Year project (FYP): “ Optimum Aggregate Filtration For Construction Site Runoff ”.
- Awarded merit for FYP and project featured in the Land Transport Authority (LTA) Bulletin, December 2010.

HONOURS AND AWARDS

Nominated for outstanding postgraduate year 1 (PGY1) in Internal Medicine

Feb 2023

- Nominated nationally as an outstanding houseman/intern for exceeding expectations as a PGY1 beyond the performance of peers, based on clinical skills, medical knowledge, communication skills, professionalism, responsibility, and teamwork.

Pre-Employment Grant (Medicine)

Nov 2020

- Awarded by Ministry of Health Holdings (MOHH) to encourage good performing students studying Medicine outside of Singapore to work as a clinician in public healthcare clusters in Singapore after graduation.

Nicholas Hang Bao Han

- Silver Award of Service by BP International Mentoring and Tutoring** Aug 2015
- Conferred for serving the student community for two years through mentoring and tutoring and completing a milestone of 200 hours of community service.
- Bronze Award for service by BP International Mentoring and Tutoring** Aug 2014
- Conferred for serving the student community for one year through mentoring and tutoring and completing a milestone of 100 hours of community service.
- Earned Band 1 (Distinction) in Safety and Repair Military Tank Audit** Dec 2011
- Recognized as part of a 10-member team for surpassing vehicle repair, maintenance, and workshop safety standards in accordance with ISO standards.
- 3rd Place for Safety Board Competition at Kranji Military Camp III** Aug 2011
- Conferred in an interdivision camp competition for the design of a safety and hazard awareness board at the workshop.
- Half Colours Award for Student Excellence** Oct 2010
- Conferred for representing NP and securing 2nd place in national international chess competition.
- Gold Certificate of Achievement in Co-curricular activities (CCA)** Oct 2010
- Conferred for outstanding student leadership and commitment in CCA over a span of 3 years in NP.
- Book prize for module “Ecology”** Oct 2007
- Awarded for best performance in the cohort in Academic Session 2007/2008
- Book prize for module “Individual and Community”** Oct 2007
- Awarded for best performance in the cohort in Academic Session 2007/2008

CURRENT/PREVIOUS CERTIFICATIONS AND ACCREDITATION

- Certified Military Tank Mechanic** May 2011
- Specialised in maintenance and repair of Bronco All-Terrain Tracked Carrier (ATTC) and Bionix II military vehicles to ensure mechanical reliability and operational readiness in field exercises.
- Workplace Safety and Health (WSH) Officer** Aug 2009
Accredited training provider
- Conduct risk identification and assessment to identify workplace hazards.
 - Manage accident reporting and instigation processes.
 - Perform job safety analysis and implemented preventive and control measures to enhance workplace safety.
- Noise Monitoring Officer** Feb 2009
Accredited training provider
- Utilise design tools to monitor noise levels and assess their impact on health
 - Implement effective preventive and control measures to manage noise-related risks.

JOURNAL PUBLICATIONS

Kihan Yap [et al, including Hang, N] “Impact of COVID-19 pandemic on the implementation of pathogen-specific transmission- based precautions” – *Manuscript under review*.

Nicholas Hang Bao Han, Irfan Sagir Khan, Bunde Manish Mahadeorao “Abortive squamous differentiation in SMARCB1 (INI1) deficient sinonasal carcinoma: potential pitfall in small biopsies versus squamous cell carcinoma with basaloid features” - *Manuscript in preparation*.

Cheng Li, Andrew L Foster , Nicholas Hang Bao Han , Andrej Trampuz , Michael Schuetz.(2022). “A bibliometric analysis of clinical research on fracture-related Infection”. *Biomed Res Int*. 2022 Apr 14;2022:8171831. doi: 10.1155/2022/8171831. PMID: 35463982; PMCID: PMC9023151.

Ashokan, A., Lim, J., Hang, N., & Mitra, R. (2018). "Complex housing causes a robust increase in dendritic complexity and spine density of medial prefrontal cortical neurons". *Nature - Scientific Reports*, 8(1). doi: 10.1038/s41598-018-25399-4.

CONFERENCE AND/OR POSTER PRESENTATIONS

Nicholas Hang Bao Han, Albert Ty, Marimuthu Kalisvar, **Methicillin-Resistant Staphylococcus Aureus (MRSA) Blood Stream Infections (BSI) control strategies in Singapore**. Second meeting of the ASEAN-US Infection Prevention and Control Task force, July 2024, Jakarta, Indonesia.

Nicholas Hang Bao Han, **Short Environmental Enrichment Rescues Effects of Chronic Stress in Infralimbic Neurons of mPFC in Adult Male Rats**. Final Year Project Poster Presentation, May 2016, NTU, Singapore

Milindu Chanaka Makandura, Nicholas Bao Han Hang, Yu Han Chee, **Tibial Plateau Fracture: Outcome and Predictors of Severity. Experience of An Orthopaedic Trauma Centre**. 16th EFORT Congress Prague, May 2015, Czech Republic.

Bryan Gan Qing Xian, Neo Wei Yi, Nicholas Hang Bao Han, **Optimum Aggregate Filtration For Construction Site Runoff**. Seminar on Earth Control Measure, IES Academy, May 2010, Singapore.

RESEARCH EXPERIENCE

Jamieson Trauma Institute (JTI) – Australia, Brisbane
Extracurricular research

Apr 2020 - Jun 2021

- Analysed and profiled global research on orthopaedic fracture-related infections over a 20-year period, identifying trends, key focus areas, and gaps in clinical and epidemiological studies.
- Conducted a bibliometric analysis to evaluate the volume, quality, and geographic distribution of research publications, highlighting global contributions to fracture-related infection management.
- Collaborated with an interdisciplinary team to synthesize findings into actionable insights that inform clinical practices and public health strategies for infection prevention.

The University of Queensland (UQ) – Australia, Brisbane
Curriculum research

Feb 2018 - Sept 2019

- Performed a scoping review of public health strategies in dengue endemic countries over a 5 year period.
- Reviewed a wide range of academic and policy literature to identify key approaches, including vector control, public awareness campaigns, and vaccination strategies.
- Synthesized findings into a comprehensive report highlighting gaps in current strategies and recommending improvements for public health initiatives.

Nicholas Hang Bao Han

- Developed skills in literature review, critical appraisal, and public health strategy evaluation.

The Queensland University of Technology (QUT) – Australia, Brisbane May 2018 – Jun 2019
Extracurricular research

- Conducted laboratory experiments to assess the antimicrobial properties of various nanoparticles against biofilm-forming pathogens commonly found in implant-related infections.

Nanyang Technological University – Singapore Dec 2015 - May 2016
Final Year Project

- Investigated effects of environmental enrichment (EE) on Infralimbic (IL) neurons in tandem with effects of chronic stress on IL neurons.
- Conducted experiments to analyze synaptic plasticity and neuronal morphology, exploring the interplay between positive environmental stimuli and stress factors on brain health.

Nanyang Technological University – Singapore May 2015 - Aug 2015
Summer research experience

- Investigated the synaptic plasticity of hippocampus of post-partum rats under chronic stress.
- Utilized advanced neuroanatomical techniques, including histological staining and microscopy, to assess changes in neuronal complexity and connectivity.

Ngee Ann Polytechnic – Singapore Oct 2010
Extracurricular research

- Collected air samples from various expressways and highways in Singapore during peak and off-peak periods over a 4-week span.
- Analysed and investigated different types and concentrations of air pollutants at various times of day to support policy drafting, in collaboration with the National Environment Agency (NEA).

Ngee Ann Polytechnic – Singapore Aug 2009 – Feb 2010
Final Year Project

- Designed and implemented an aggregate filtration system to minimize sediment runoff from construction sites, reducing environmental pollution.
- Collaborated with team to develop sustainable solutions for sediment management in construction settings, based on environmental engineering principles.
- Conducted an extensive literature review on existing practices and challenges, informing project design and implementation.
 Presented findings and project outcomes to LTA, showcasing technical expertise and practical applicability.

PROFESSIONAL EXPERIENCE/ WORK AND PLACEMENTS

Ministry of Health (MOH), Singapore Jul 2024 – Current
Medical Officer, (Non-Communicable) Disease Policy & Strategy Division (DPSP)

- Collaborated with interdisciplinary teams within MOH and external agencies (e.g. Ministry of Sustainability and the Environment, National Environment Agency) to shape national health policies addressing environmental challenges and rising temperatures, including the strategic design of Research and Development (R&D) roadmaps for heat resilience.
- Co-draft heat wave emergency contingency protocols in partnership with various ministries and agencies.
- Create a standardised criteria for reporting heat-related injuries and fatalities, collaborating with an interdisciplinary team within MOH.
- Developed public-facing clinical information to increase awareness of heat-related health risks and preventive measures.

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- Co-draft the national medical toxins response plan with toxicologists and crisis response experts, contributing to Singapore's disaster readiness.
- Conduct comprehensive literature reviews on the effects of cosmic radiation on airplane crewmembers and on noise induced hearing loss in Singapore.
- Revise screening guidelines for 65 medical conditions (e.g. diabetes, hypertension and osteoporosis) in collaboration with the Screening Test Review Committee (STRC) and Saw Swee Hock School of Public Health (SSHSPH).
- Draft project charters [e.g. nasopharyngeal cancer and spinal muscular atrophy (SMA)] for literature review on medical screening tests in collaboration with the public health translational team at SSHSPH.
- Review and appraise scientific evidence for various medical screening tests and drafting recommendations to the STRC and Academy of Medicine Singapore (AMS) for guidelines endorsement.
- Collaborate with interdisciplinary teams within MOH and external agencies to formulate and implement chronic national health policies in the Singapore context.
- Collaborate on a national developmental needs prevalence study for children aged 0-6 with MOH, SSHSPH, Early Childhood Developmental Agency (ECDA) and clinicians
- Evaluate data from the Singapore National Birth Defects Registry and assess the feasibility of decommissioning the registry.
- Conceptualise and create a precision medicine genetic screening and diagnostic framework with an interdisciplinary team within MOH to establish a national standard for genetic testing protocols.
- Drafted responses to parliamentary questions on public health issues (e.g. hearing loss trends, childhood allergy rates, menopausal statistics).
- Draft official media statements addressing sensitive health topics (e.g. fluoride use in drinking water) to ensure accurate public information and communication.
- Assist in ad hoc consultancy projects with MOH and external agencies, including rapid reviews on international screening programs and a 10-year analysis of heat-related injuries in Singapore.
- Consult on international health projects, such as advising the Ministry of Health, Saudi Arabia, on streamlining their death registration process.

Ministry of Health (MOH), Singapore

Jan 2024 – Jul 2024

Medical Officer, Patient safety and quality improvement division (PSQI)

- Developed and implemented Infection, Prevention and Control (IPC) health policies in collaboration with MOH and healthcare institutes, improving infection control practices nationwide.
- Analysed and published annual national IPC indicators reports to provide critical insights and feedback to public hospitals on their IPC matters.
- Co-drafted and revised multiple IPC guidelines and standards in partnership the National Infection Prevention and Control (NIPC) Committee to ensure up to date best practices on IPC measures across healthcare facilities nationally.
- Analysed nationwide COVID-19 and Carbapenemase-Producing Enterobacteriaceae (CPE) hospital surveys to guide national surveillance efforts.
- Drafted a national circular for acute tertiary hospitals to strengthen CPE surveillance and control measures.
- Drafted the NIPC Committee's three-year work plan, streamlining efforts and improving focus.
- Developed a proposal for a healthcare-associated infection point prevalence study, designed to comprehensively assess the prevalence of such infections in community hospitals.
- Co-drafted a grant proposal with a budget of approximately 122,000,00 SGD (~91,000 USD) for a study on community-onset Methicillin-resistant *Staphylococcus aureus* bloodstream infections.
- Educated Brunei delegates on Singapore's IPC practices during their intercountry visit, fostering regional collaboration and knowledge exchange.

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- Planned and coordinated monthly NIPC committee meetings, including agenda setting and drafting meeting minutes.
- Collaborated with external partners and agencies on IPC initiatives to reinforce national IPC standards.
- Supported the Association of Southeast Asian Nations (ASEAN) -United States IPC Taskforce in enhancing IPC programs across Southeast Asia (e.g. regional IPC project planning).
- Provided expertise on ad hoc IPC consultancy projects, such as ICU ventilation design for a new hospital (Eastern General Hospital) and implementation of spot cooling measures to address hospital heat stress.
- Provided multiple clinical inputs for root cause analyses of serious adverse events in hospital and clinic settings, identifying iatrogenic risk factors (e.g. hepatitis C exposure) and contributing to risk reduction efforts.
- Developed corrective action plans to address risks and enhance hospital processes.
- Conducted priority setting and established a cross-divisional workgroup within MOH to advance initiatives at the MOH Office for Healthcare Transformation (MOHT).
- Co-drafted a speech for the Senior Minister of State, Dr Janil Puthuchear at the SingHealth Inspirational Patient and Caregiver Awards Ceremony 2024, recognizing and celebrating resilience and contributions of patients and caregivers.

Forensic Pathology, Health Sciences Authority (HSA)

Jul 2023 – Dec 2023

Medical Officer

- Clerked daily coroner cases with team consultants to evaluate the necessity of autopsies.
- Performed charting and examinations for non-autopsy coroner cases, contributing to external assessment and investigation.
- Conducted over 80 autopsies, determining the cause and manner of death in a diverse range of cases.
- Interpreted postmortem toxicology and histopathology results, integrating findings to support accurate cause-of-death conclusions.
- Drafted comprehensive medicolegal autopsy reports, ensuring precise and detailed documentation for legal and investigative use.
- Liaised with the Singapore Police Force (SPF), legal professionals, and other agencies, drafting correspondences and facilitating communication regarding forensic investigations.

Anatomical Pathology, Singapore General Hospital (SGH)

May 2023 – Jul 2023

Medical Officer

- Trimmed over 100 hospital specimens, performing macroscopic examination to identify tumours or abnormalities.
- Assisted in diagnosing and prognosticating disease conditions by analysing immunohistochemistry and microscopic histological findings.
- Drafted detailed biopsy specimen reports, ensuring accurate and comprehensive documentation of findings.
- Rotated across various departments and laboratories, including cytology, frozen section, electron microscopy, tissue processing and fixation, and haematology, to gain a broad understanding of work processes and techniques.

PGY1 Hospital internship/ Singapore

Mar 2022 - Feb 2023

House Officer

- Rotated through general medicine, surgery, and subspecialty departments across multiple hospitals in Singapore, including Teng Fong General Hospital (NTFGH), Singapore General Hospital (SGH), and Tan Tock Seng Hospital (TTSH), gaining broad clinical experience.
- Reviewed patient cases and developed treatment plans, providing direct patient care under the supervision of senior medical staff.
- Collaborated with allied health professionals to create and implement comprehensive patient treatment plans, ensuring continuity of care both in-hospital and during post-discharge follow-up.

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- Liaised with patients' next of kin (NOK), providing regular updates and support to family members throughout the patient's hospital stay.
- Actively participated in onsite learning sessions to enhance medical knowledge and skills as part of continuing professional development.

Mount Elizabeth Hospital, Singapore

Dec 2018 – Jan 2019

Medical Observership, Ear Nose Throat (ENT) clinic

- Observed and assisted in various ENT procedures such as endoscopies, biopsies, and minor surgical interventions.
- Observed and assisted in audiological assessments and treatments, including hearing tests, hearing aid fittings, and auditory rehabilitation.

Dhulikhel Hospital, Nepal, Kathmandu

Nov 2018 - Dec 2018

Medical Observership

- Assisted consultants with diagnosis and management during clinical sessions in the Orthopaedic and Dermatology departments, gaining valuable insights into patient care and treatment.

Adept Health Pte Ltd, Singapore

Jun 2017 - Dec 2017

General Administrator

- Completed claims paperwork, documentation, and system entries accurately and efficiently.
- Identified and corrected inconsistencies and discrepancies in medical documentation, ensuring data integrity.
- Verified eligibility and claims status for thousands of patients with insurance agencies, facilitating smooth processing and resolution of claims.

Quest Laboratories Pte Ltd, Singapore

Jun 2014 - July 2014

Medical Technologist Intern

- Prepared specimens for analysis, establishing appropriate priorities for testing to ensure timely and accurate results.
- Identified abnormal test results and referred them to Clinical Laboratory Scientists for further review and follow-up.
- Reviewed test results meticulously for accuracy, contributing to reliable diagnostic outcomes.
- Performed diagnostic and therapeutic laboratory procedures, gaining hands-on experience in laboratory practices and techniques.

Singapore Armed Forces (SAF) – REC

Nov 2010 - Aug 2012

Military Tank Mechanic, Workshop Safety Officer

- Collaborated in teams of 4 to 7 members to repair and replace damaged tank parts, ensuring effective team coordination and problem-solving.
- Served as a designated workshop safety officer, overseeing adherence to safety protocols and promoting a secure work environment.
- Assisted in annual safety audits of military tanks, contributing to compliance and safety standards.
- Repaired defects in basic cooling systems, drive systems, steering, and suspension systems of military tanks.
- Performed daily maintenance and routine vehicle checks, including fluid levels, hoses, belts, brakes, tires, and clutches, to ensure operational readiness.
- Implemented workshop safety protocols, enhancing safety and efficiency in the maintenance environment.
- Assisted in annual safety audits of military tanks.
- Troubleshoot issues with tanks during repair services, resolving mechanical problems efficiently.

TÜV SÜD PSB Accreditation and Certification Agency, Singapore
Student Internship, Laboratory Officer and Environmental Surveyor

Mar 2009 – May 2009

- Conducted laboratory tests and environmental surveys to ensure compliance with industry standards, including metal toxicology and air quality testing in factory settings.
- Analysed and interpreted laboratory and environmental test results, supporting certification and accreditation processes.
- Performed on-site inspections, sample collection (e.g., cooling water towers, air handling units in buildings), and workplace safety audits to ensure regulatory compliance.
- Provided technical advisory support, offering insights on occupational and environmental standards and safety practices.
- Contributed to the preparation of comprehensive reports and documentation for environmental assessments.
- Acquired practical experience in laboratory procedures and environmental inspection techniques.

BreadTalk Group Private Limited, Singapore
Food and Beverage Cashier

Mar 2007 – Apr 2007

- Handled and maintained accurate cash register records, balanced cash drawers at the end of shifts, and reported discrepancies to management
- Assisted in stocking and organizing merchandise, ensuring shelves were well-stocked and products were properly displayed.
- Collaborated with team members to meet sales goals and uphold high standards of store presentation and cleanliness.
- Supported promotional activities and upselling opportunities to drive sales and enhance the customer experience.
- Provided quality customer service, addressing customer inquiries and resolving issues to ensure a positive shopping experience.

LEADERSHIP, TEACHING AND CO-CURRICULUM ACTIVITIES

Educational Talk on Sexually Transmitted Infections (STIs)

One World International School (OWIS), Guest speaker

24 May 2024

- Educated 150 Grade 7 & 8 students on STIs in accordance to the personal, social, health and economic education (PSHE) curriculum.
- Developed and presented educational materials tailored to the age group and cultural context of the students.
- Engaged with students through interactive discussions and Q&A sessions to enhance understanding and address concerns.

Singapore medical association of Queensland (SMAQ) committee member
Treasurer and Secretary

Oct 2019 - Oct 2020

- Administered a budget of 2000 AUD (~1300 USD) for the club, allocating funds for five major events throughout the year.
- Handled administrative tasks with club operations and external collaborators.
- Oversaw and maintained a study resources database for medical students

Global Medical Project India, Odisha
Event planner and coordinator

Nov 2019 - Dec 2019

- Organised and executed games and educational events for 200 orphanage children, including Cardio pulmonary resuscitation (CPR) and hand hygiene sessions.
- Coordinated a fundraiser that raised 5000 AUD (~3250 USD) to purchase Tetanus and HPV vaccines for the children.
- Liaised with Indian pharmaceutical companies for vaccine procurement.

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- Administered vaccines, and performed basic medical check-ups for 200 orphanage children alongside a medical practitioner.
- Provided medical supplies and assisted in check-ups for villagers in remote parts of Odisha, India.

The University of Queensland Medical Society (UQMS) Peer Tutoring Student Tutor Jan 2019 - Dec 2019

- Instructed seven first-year medical students in relevant subjects on a weekly basis.
- Developed and planned tutorial sessions and educational materials for students.

NTU student union BP mentoring Programme, Committee Member Financial Controller Jan 2014 - Dec 2014

- Managed a treasury of 2000 SGD (~1490 USD) for the BP Mentoring Programme, budgeting for events and liaising with BP for funding.
- Provided weekly mentoring to 3-4 secondary school students in Science and Mathematics.

NTU student union BP mentoring programme, Committee Member Liaison Director Jan 2013 - Dec 2013

- Liaised with schools participating in the mentoring programme and coordinated weekly sessions.
- Drafted weekly reports and managed 12 mentors for each session.
- Tutored 4-5 secondary school students in Science and Mathematics and planned weekly mentoring events.

NTU Inter- Hall recreational competition, Committee Member Chinese-Chess Captain Dec 2013 - Jan 2014

- Led and organized weekly training sessions for team members.

NTU Inter- Hall recreational competition, Member International Chess Representative Dec 2013 - Jan 2014

- Achieved 1st place in the inter-hall recreational games.

Choir NTU Tenor Singer Aug 2013 - Oct 2014

- Participated in various performances and events.

NP Chess club, Committee member International Chess Captain Apr 2008 - May 2010

- Organised regular training sessions and collaborated with chess coaches to improve skill and strategy of team.
- Planned and executed 3 school events and 2 internal competitions.
- Achieved 2nd place in interschool International Chess competition.

NP Green Volunteers Committee Member (2008) Apr 2007 – Oct 2010

- Co-coordinated and co-planned two weekly clean and green campaign events, promoting environmental awareness and action within the school.

Singapore Taekwondo Federation Black Tip Apr 2003 - Oct 2005

COMMUNITY SERVICE

Matchsticks of IMH, Institute of mental health (IMH), Singapore Ward Leader Jan 2017 - Dec 2017

Nicholas Hang Bao Han

- Supervised and coordinated a team of 8-10 volunteers in ward sessions, ensuring effective support and engagement.
- Provided social and emotional support to long-term stay patients, enhancing their well-being and quality of life.
- Organized and planned events, entertainment, and celebrations to improve patient morale and foster a supportive environment.

Friends of NCIS, National Cancer Institute of Singapore (NCIS)

Oct 2015 - May 2016

Volunteer

- Offered social support to cancer patients undergoing treatment, providing comfort and companionship.
- Facilitated activities for 20 cancer survivors and their support groups, promoting community and mutual support.
- Contributed to cancer awareness by participating in two public education campaigns, enhancing community understanding of cancer.

Sabah Community Service, Youth Expedition Project, Malaysia

Mar 2008

Volunteer

- Assisted in the installation of a standardised clean water supply system for local village communities, improving access to safe water.
- Taught English and basic computing skills to 20 children, supporting their education and skill development.
- Managed the procurement and preparation of cooking ingredients for meals for a youth expedition team of 15 members for 2 weeks.

LANGUAGES

- English: Native Proficiency
- Mandarin Chinese: Professional Proficiency
- Japanese: Intermediate Proficiency
- Korean: Basic Proficiency

TRAITS, SKILLS AND HOBBIES

Traits

- Critical thinker
- Adaptable learner
- Quality-focused

Skills

- Proficient in Microsoft Word, PowerPoint and Excel.
- Competent in Statistical Package for the Social Sciences (SPSS) software.

Hobbies

- Learning languages
- Swimming
- Bouldering